CANONICAL MAP FROM A COMPLEX OF MODULES TO THE DERIVED PUSHFORWARD, FUNCTORIAL IN THE TRIPLE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The 'moreover' clause on functoriality is discharged by the author through the construction itself; no separate argument is supplied in the source and none was added.

Editorial note. The preamble is the chapter's retained minimal preamble (sources/stacks-cohomology/retained-preamble.tex lines 1-81), a verbatim excerpt of the native preamble: Xy-pic setup plus the native theorem declarations and mathematical macros.

Editorial note. The label lemma-functoriality is defined only in cohomology.tex, so no cross-chapter restatement had to be weighed for this card.

Allowed standard prerequisites (editorial).

- Injective resolutions in the category of O_X-modules. Same chapter, the existence of enough injectives among O_X-modules, used implicitly at proof line 2348. Supplies the resolution F to I from which the representative f_*I of Rf_*F is built.
- The derived category D^+(Y) as a localization of K^+(Y). Homology, the construction of D(A) and the localization functor j_Y: K^+(Y) to D^+(Y) named at proof line 2357. Supplies the last step of the construction and the meaning of a canonical morphism in D^+(Y).
- Definition of the derived pushforward Rf_*. Same chapter, Rf_* computed on an injective resolution, used at proof lines 2349-2350. Supplies that f_*I represents Rf_*F in K^+(O_Y).

Lemma 0.1. *Let $f : X \to Y$ be a morphism of ringed spaces. Let $\mathcal{G}^\bullet$, resp. $\mathcal{F}^\bullet$ be a bounded below complex of $\mathcal{O}_Y$-modules, resp. $\mathcal{O}_X$-modules. Let $\varphi : \mathcal{G}^\bullet \to f_*\mathcal{F}^\bullet$ be a morphism of complexes. There is a canonical morphism*

$$\mathcal{G}^\bullet \longrightarrow Rf_*(\mathcal{F}^\bullet)$$

in $D^+(Y)$. Moreover this construction is functorial in the triple $(\mathcal{G}^\bullet, \mathcal{F}^\bullet, \varphi)$.

Proof. Choose an injective resolution $\mathcal{F}^\bullet \to \mathcal{I}^\bullet$. By definition $Rf_*(\mathcal{F}^\bullet)$ is represented by $f_*\mathcal{I}^\bullet$ in $K^+(\mathcal{O}_Y)$. The composition

$$\mathcal{G}^\bullet \to f_*\mathcal{F}^\bullet \to f_*\mathcal{I}^\bullet$$

is a morphism in $K^+(Y)$ which turns into the morphism of the lemma upon applying the localization functor $j_Y : K^+(Y) \to D^+(Y)$. □

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